2501.06662: equation evidence

Display equations

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>2501.06662_EQ0051</td><td>21</td><td>■ 0.536</td><td><code>\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t} . \end{aligned}</code></td><td>$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} . \end{aligned}$</td><td>$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text { nondeg. paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}} \text{ in } \mathsf{L} \quad (-1)^{k} \prod_{i=1}^{k} \pi(y_{i}|y_{i-1})^{t}. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0059</td><td>23</td><td>■ 0.550</td><td><code>\begin{aligned} \lim _{n \rightarrow \infty} & \frac{ \mathscr{H}\left(\mathcal{M},\left.\pi(-\mid \perp) \right|_{T(\perp)}, \zeta\right)}{n} \\ & =\lim _{n \rightarrow \infty}-\frac{1}{n} \sum_{\left(a_{1}, \ldots, a_{N-1}\right) \in A^{N-1}} \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \log \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \end{aligned}</code></td><td>$\begin{aligned} \lim _{n \rightarrow \infty} & \frac{\mathscr{H}\left(\mathcal{M},\left.\pi(-\mid \perp)\right|_{T(\perp)}, \zeta\right)}{n} \\ & =\lim _{n \rightarrow \infty}-\frac{1}{n} \sum_{\left(a_{1}, \ldots, a_{N-1}\right) \in A^{N-1}} \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \log \pi\left(\perp a_{1} \cdots a_{N-1} \mid \perp\right) \end{aligned}$</td><td>$\begin{aligned} \lim _{n \rightarrow \infty} & \frac{\mathscr{H}(\mathcal{M},\pi(-|\perp)|_{T(\perp)}, \zeta)}{n} \\ & =\lim _{n \rightarrow \infty}-\frac{1}{n} \sum_{(a_{1}, \ldots, a_{N-1}) \in A^{N-1}} \pi(\perp a_{1} \cdots a_{N-1}|\perp) \log \pi(\perp a_{1} \cdots a_{N-1}|\perp) \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0058</td><td>23</td><td>■ 0.594</td><td><code>\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \zeta_{t}(y, z) p(z)\right) \\ & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \pi(z \mid y)^{t} p(z)\right) . \end{aligned}</code></td><td>$\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \zeta_{t}(y, z) p(z)\right) \\ & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \pi(z \mid y)^{t} p(z)\right) . \end{aligned}$</td><td>$\begin{aligned} \mathscr{H}(\mathcal{M}, p, \zeta) & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \zeta_{t}(y, z) p(z)\right) \\ & =-\sum_{y \in \mathrm{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \mathrm{ob}(\mathcal{M})} \pi(z|y)^{t} p(z)\right) . \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0023</td><td>15</td><td>■ 0.636</td><td><code>\begin{aligned} \#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i} \right\} & (\# A+1)=\left(\sum_{i=0}^{N-2} \# A^{i} \right)(\# A+1) \\ & =\# A^{N-1}+2\left(\# A^{N-2} \right)+\cdots+2(\# A)+1 \end{aligned}</code></td><td>$\begin{aligned} \#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\} & (\# A+1)=\left(\sum_{i=0}^{N-2} \# A^{i}\right)(\# A+1) \\ & =\# A^{N-1}+2\left(\# A^{N-2}\right)+\cdots+2(\# A)+1 \end{aligned}$</td><td>$\begin{aligned} \#\left\{x \in \mathrm{ob}(\mathsf{L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\} (\# A+1) & =\left(\sum_{i=0}^{N-2} \# A^{i}\right)(\# A+1) \\ & =\# A^{N-1}+2(\# A^{N-2})+\cdots+2(\# A)+1. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0052 (13)</td><td>21</td><td>■ 0.693</td><td><code>\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{ \ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell} .</code></td><td>$\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{\ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell} .$</td><td>$\operatorname{Mag}(t \mathcal{M})=\sum_{k \geq 0} \sum_{\ell \in[0, \infty)} \sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell} .$</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_EQ0033	17	■0.720	`\zeta_{t}^{-1}(x, y)= \begin{cases}-\pi(y \mid x)^{t} & \text { if } y \in \mathrm{~L}_{x}^{(1)} \\ 1 & \text { if } y=x \\ 0 & \text { otherwise }\end{cases}`	$\zeta_{t}^{-1}(x, y)= \begin{cases}-\pi(y \mid x)^{t} & \text { if } y \in \mathrm{~L}_{x}^{(1)} \\ 1 & \text { if } y=x \\ 0 & \text { otherwise }\end{cases}$	$\zeta_t^{-1}(x,y)=\begin{cases}-\pi(y\mid x)^t & \textit{if } y\in \mathsf{L}_x^{(1)} \\ 1 & \textit{if } y=x \\ 0 & \textit{otherwise.}\end{cases}$
2501.06662_EQ0037 (10)	18	■0.780	`\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, x)-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} \sum_{y \in \mathrm{~L}_{x}^{(1)}} \pi(y \mid x)^{t} \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)}\left(1-\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}\right)+\#(T(\perp)) . \end{aligned}`	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) & =\sum_{x, y \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, x)-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} \sum_{y \in \mathrm{~L}_{x}^{(1)}} \pi(y \mid x)^{t} \\ & =\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)}\left(1-\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}\right)+\#(T(\perp)) . \end{aligned}$	$\begin{aligned}\operatorname{Mag}(t\mathcal{M}) &= \sum_{x,y\in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,y) \\ &= \sum_{x\in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,x) - \sum_{x\in \mathrm{ob}(\mathcal{M})\backslash T(\perp)} \sum_{y\in \mathsf{L}_x^{(1)}} \pi(y\mid x)^t \\ &= \sum_{x\in \mathrm{ob}(\mathcal{M})\backslash T(\perp)} \left(1-\sum_{a\in A\cup\{\dagger\}} p_x(a)^t\right)+\#(T(\perp)).\end{aligned}$
2501.06662_EQ0049	20	■0.797	`\begin{aligned} & \partial_{k}^{i}\left(y_{0}, \ldots, y_{k}\right)= \\ & \begin{cases}\left(y_{0}, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{k}\right) & \text { if } d\left(y_{i-1}, y_{i}\right)+d\left(y_{i}, y_{i+1} \right)=d\left(y_{i-1}, y_{i+1}\right) \\ 0 & \text { otherwise }\end{cases} \end{aligned}`	$\begin{aligned} & \partial_{k}^{i}\left(y_{0}, \ldots, y_{k}\right)= \\ & \begin{cases}\left(y_{0}, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{k}\right) & \text { if } d\left(y_{i-1}, y_{i}\right)+d\left(y_{i}, y_{i+1} \right)=d\left(y_{i-1}, y_{i+1}\right) \\ 0 & \text { otherwise }\end{cases} \end{aligned}$	$\begin{aligned}&\partial_k^i(y_0,\ldots,y_k)= \\ &\qquad \begin{cases}(y_0,\ldots,y_{i-1},y_{i+1},\ldots,y_k) & \text{if } d(y_{i-1},y_i)+d(y_i,y_{i+1})=d(y_{i-1},y_{i+1}) \\ 0 & \text{otherwise}\end{cases}\end{aligned}$
2501.06662_EQ0001	3	■0.822	`\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .`	$\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x\in \mathrm{ob}(\mathcal{M})\backslash T(\perp)} H_t(p_x) + \#(T(\perp)), \qquad t>0.$
2501.06662_EQ0029	16	■0.829	`\left(\zeta_{t}-\delta\right)^{k}(x, y)=\sum_{\substack{ \left(y_{0}, y_{1}, \ldots, y_{k-1}, y_{k}\right) \\ y_{0}=x, y_{k}=y \\\left(y_{i-1}, y_{i}\right) \in E \text { for } i=1, \ldots, k}} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t}`	$\left(\zeta_{t}-\delta\right)^{k}(x, y)=\sum_{\substack{ \left(y_{0}, y_{1}, \ldots, y_{k-1}, y_{k}\right) \\ y_{0}=x, y_{k}=y \\\left(y_{i-1}, y_{i}\right) \in E \text { for } i=1, \ldots, k}} \prod_{i=1}^{k} \pi \left(y_{i} \mid y_{i-1}\right)^{t}$	$(\zeta_t-\delta)^k(x,y) = \sum_{\substack{(y_0,y_1,\ldots,y_{k-1},y_k) \\ y_0=x,\, y_k=y \\ (y_{i-1},y_i)\in E \text{ for } i=1,\ldots,k}} \prod_{i=1}^{k} \pi(y_i\mid y_{i-1})^t.$
2501.06662_EQ0036 (11)	18	■0.835	`\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .`	$\operatorname{Mag}(t \mathcal{M})=(t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H_{t} \left(p_{x}\right)+\#(T(\perp)), \quad t>0 .$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x\in \mathrm{ob}(\mathcal{M})\backslash T(\perp)} H_t(p_x) + \#(T(\perp)), \qquad t>0.$

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<tr><td>2501.06662_EQ0026 (8)</td><td>15</td><td>■ 0.842</td><td><code>\zeta_{t}^{-1}(x, y)=\sum_{k \geq 0} \sum_{\substack{ \text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} .</code></td><td>$\zeta_{t}^{-1}(x, y)=\sum_{k \geq 0} \sum_{\substack{ \text { nondeg.paths } \\ x=y_{0} \rightarrow y_{1} \rightarrow \cdots \rightarrow y_{k}=y}}(-1)^{k} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} .$</td><td>$\zeta_t^{-1}(x,y)=\sum_{k\geq 0}\sum_{\substack{\textit{nondeg. paths}\\ x=y_0\to y_1\to\cdots\to y_k=y}}{}_{\textit{in } \mathsf{L}}(-1)^k\prod_{i=1}^{k}\pi(y_i|y_{i-1})^t.$</td></tr>
<tr><td>2501.06662_EQ0021</td><td>14</td><td>■ 0.849</td><td><code>\begin{aligned} \operatorname{Mag}(\mathrm{L}) & = \sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \zeta_{ \mathrm{L}}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, x)+ \sum_{\substack{x=\perp a \in \mathrm{ob}(\mathrm{~L}) \\ a \in \bigcup_{i=0}^{N-2} A^{i}}} \sum_{a_{1} \in A \cup\{\dagger\}} \mu_{\mathrm{L}}\left(x, x a_{1} \right) \\ & =\# \mathrm{ob}(\mathrm{~L})-\#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\}(\# A+1) \end{aligned}</code></td><td>$\begin{aligned} \operatorname{Mag}(\mathrm{L}) & = \sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \zeta_{ \mathrm{L}}^{-1}(x, y) \\ & =\sum_{x, y \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, y) \\ & =\sum_{x \in \mathrm{ob}(\mathrm{~L})} \mu_{\mathrm{L}}(x, x)+ \sum_{\substack{x=\perp a \in \mathrm{ob}(\mathrm{~L}) \\ a \in \bigcup_{i=0}^{N-2} A^{i}}} \sum_{a_{1} \in A \cup\{\dagger\}} \mu_{\mathrm{L}}\left(x, x a_{1} \right) \\ & =\# \mathrm{ob}(\mathrm{~L})-\#\left\{x \in \mathrm{ob}(\mathrm{~L}): x=\perp a \text { with } a \in \bigcup_{i=0}^{N-2} A^{i}\right\}(\# A+1) \end{aligned}$</td><td>$\begin{aligned} \operatorname{Mag}(\mathsf{L}) &= \sum_{x,y\in\mathrm{ob}(\mathsf{L})}\zeta_{\mathsf{L}}^{-1}(x,y) \\ &= \sum_{x,y\in\mathrm{ob}(\mathsf{L})}\mu_{\mathsf{L}}(x,y) \\ &= \sum_{x\in\mathrm{ob}(\mathsf{L})}\mu_{\mathsf{L}}(x,x)+\sum_{\substack{x=\perp a\in\mathrm{ob}(\mathsf{L})\\ a\in\bigcup_{i=0}^{N-2}A^i}}\sum_{a_1\in A\cup\{\dagger\}}\mu_{\mathsf{L}}(x,xa_1) \\ &= \#\,\mathrm{ob}(\mathsf{L})-\#\left\{x\in\mathrm{ob}(\mathsf{L}) : x=\perp a \text{ with } a\in\bigcup_{i=0}^{N-2}A^i\right\}(\#A+1). \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0022</td><td>15</td><td>■ 0.871</td><td><code>\begin{aligned} \# \mathrm{ob}(\mathrm{~L}) & =1+ \sum_{i=1}^{N-1} \# \mathrm{~L}^{(i)} \\ & =1+ \sum_{i=1}^{N-1} \# A^{i}+\# A^{i-1}=\# A^{N-1}+2\left(\# A^{N-2}\right)+\cdots+2(\# A)+2 \end{aligned}</code></td><td>$\begin{aligned} \# \mathrm{ob}(\mathrm{~L}) & =1+ \sum_{i=1}^{N-1} \# \mathrm{~L}^{(i)} \\ & =1+ \sum_{i=1}^{N-1} \# A^{i}+\# A^{i-1}=\# A^{N-1}+2\left(\# A^{N-2}\right)+\cdots+2(\# A)+2 \end{aligned}$</td><td>$\begin{aligned} \#\,\mathrm{ob}(\mathsf{L}) &= 1+\sum_{i=1}^{N-1}\#\mathsf{L}^{(i)} \\ &= 1+\sum_{i=1}^{N-1}\#A^i+\#A^{i-1}=\#A^{N-1}+2(\#A^{N-2})+\cdots+2(\#A)+2. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0045</td><td>19</td><td>■ 0.922</td><td><code>\zeta_{\mathcal{M}_{x}}^{-1}=\left.\zeta_{ \mathcal{M}}^{-1}\right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}</code></td><td>$\zeta_{\mathcal{M}_{x}}^{-1}=\left.\zeta_{ \mathcal{M}}^{-1}\right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}$</td><td>$\zeta_{\mathcal{M}_x}^{-1}=\zeta_{\mathcal{M}}^{-1}\big|_{\mathrm{ob}(\mathcal{M}_x)\times\mathrm{ob}(\mathcal{M}_x)},$</td></tr>
<tr><td>2501.06662_EQ0039 (12)</td><td>18</td><td>■ 0.940</td><td><code>f^{\prime}(1)=\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_{x}\right) .</code></td><td>$f^{\prime}(1)=\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_{x}\right) .$</td><td>$f'(1)=\sum_{x\in\mathrm{ob}(\mathcal{M})\setminus T(\perp)}H(p_x).$</td></tr>
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<tr><td>2501.06662_EQ0046</td><td>20</td><td>0.942</td><td>\begin{aligned} \operatorname{Mag}\left(t \mathcal{M}_{x} \right) & =\sum_{y, z \in \mathrm{ob}\left(\mathcal{M}_{x}\right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, z) \\ & =\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, y)-\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x}\right) \backslash T(x)} \sum_{y \in \mathrm{~L}_{y}^{(1)}} \pi(z \mid y)^{t} \\ & =(t-1) \sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right) \backslash T(x)} H_{t}\left(p_{y}\right)+\#(T(x)) . \end{aligned}</td><td>$$\begin{aligned} \operatorname{Mag}\left(t \mathcal{M}_{x} \right) & =\sum_{y, z \in \mathrm{ob}\left(\mathcal{M}_{x}\right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, z) \\ & =\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right)} \zeta_{\mathcal{M}_{x}}^{-1}(y, y)-\sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x}\right) \backslash T(x)} \sum_{y \in \mathrm{~L}_{y}^{(1)}} \pi(z \mid y)^{t} \\ & =(t-1) \sum_{y \in \mathrm{ob}\left(\mathcal{M}_{x} \right) \backslash T(x)} H_{t}\left(p_{y}\right)+\#(T(x)) . \end{aligned}$$</td><td>$$\begin{aligned} \operatorname{Mag}(t\mathcal{M}_x) &= \sum_{y,z\in\mathrm{ob}(\mathcal{M}_x)} \zeta^{-1}_{\mathcal{M}_x}(y,z) \\ &= \sum_{y\in\mathrm{ob}(\mathcal{M}_x)} \zeta^{-1}_{\mathcal{M}_x}(y,y) - \sum_{y\in\mathrm{ob}(\mathcal{M}_x)\setminus T(x)} \sum_{y\in\mathsf{L}_y^{(1)}} \pi(z|y)^t \\ &= (t-1) \sum_{y\in\mathrm{ob}(\mathcal{M}_x)\setminus T(x)} H_t(p_y) + \#(T(x)). \end{aligned}$$</td></tr>
<tr><td>2501.06662_EQ0040</td><td>19</td><td>0.945</td><td>f(t)=\#(\mathrm{ob}(\mathcal{M}))-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} Z_{x}(t),</td><td>$$f(t)=\#(\mathrm{ob}(\mathcal{M}))-\sum_{x \in \mathrm{ob}(\mathcal{M}) \backslash T(\perp)} Z_{x}(t),$$</td><td>$$f(t) = \#(\mathrm{ob}(\mathcal{M})) - \sum_{x\in\mathrm{ob}(\mathcal{M})\setminus T(\perp)} Z_x(t),$$</td></tr>
<tr><td>2501.06662_EQ0044</td><td>19</td><td>0.971</td><td>\zeta_{\mathcal{M}_{x}}=\left.\zeta_{\mathcal{M}}\right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}</td><td>$$\zeta_{\mathcal{M}_{x}}=\left.\zeta_{\mathcal{M}}\right|_{\operatorname{ob}\left(\mathcal{M}_{x}\right) \times \operatorname{ob}\left(\mathcal{M}_{x}\right)}$$</td><td>$$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}}\big|_{\mathrm{ob}(\mathcal{M}_x)\times\mathrm{ob}(\mathcal{M}_x)}$$</td></tr>
<tr><td>2501.06662_EQ0032</td><td>17</td><td>0.973</td><td>\pi(y \mid x)^{t} \sum_{k \geq 0}(-1)^{k} \#\{\text { nondegenerate paths of length } k \text { from } x \text { to } y \text { in } \mathrm{L}\}</td><td>$$\pi(y \mid x)^{t} \sum_{k \geq 0}(-1)^{k} \#\{\text { nondegenerate paths of length } k \text { from } x \text { to } y \text { in } \mathrm{L}\}$$</td><td>$$\pi(y|x)^t \sum_{k\geq 0}(-1)^k \#\{\text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathsf{L}\}$$</td></tr>
<tr><td>2501.06662_EQ0056</td><td>22</td><td>0.986</td><td>P P L(y)=1 / \zeta_{t}\left(a_{0}, y\right)=1 / e^{t \ln \pi\left(y \mid a_{0}\right)},</td><td>$$P P L(y)=1 / \zeta_{t}\left(a_{0}, y\right)=1 / e^{t \ln \pi\left(y \mid a_{0}\right)},$$</td><td>$$PPL(y) = 1/\zeta_t(a_0,y) = 1/e^{t\ln\pi(y|a_0)},$$</td></tr>
<tr><td>2501.06662_ EQ0015 (7)</td><td>12</td><td>0.988</td><td>\operatorname{Mag}(\mathcal{C})=\sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y) .</td><td>$$\operatorname{Mag}(\mathcal{C})=\sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y) .$$</td><td>$$\operatorname{Mag}(\mathcal{C}) = \sum_{(x,y)\in\mathrm{ob}(\mathcal{C})\times\mathrm{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x,y).$$</td></tr>
<tr><td>2501.06662_EQ0002</td><td>4</td><td>0.991</td><td>\frac{(t-1)\left(n^{1-t}-1\right)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+ \#(T(\perp))</td><td>$$\frac{(t-1)\left(n^{1-t}-1\right)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+ \#(T(\perp))$$</td><td>$$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\mathrm{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$$</td></tr>
<tr><td>2501.06662_EQ0005</td><td>6</td><td>0.991</td><td>\operatorname{ob}(\mathrm{L})=\left\{\perp a: a \in A^{*} \text { and }|a| \leq N-1\right\} \sqcup\left\{\perp a \dagger: a \in A^{*} \text { and }|a|<N-1\right\} .</td><td>$$\operatorname{ob}(\mathrm{L})=\left\{\perp a: a \in A^{*} \text { and }|a| \leq N-1\right\} \sqcup\left\{\perp a \dagger: a \in A^{*} \text { and }|a|<N-1\right\} .$$</td><td>$$\mathrm{ob}(\mathsf{L}) = \{\perp a : a \in A^* \text{ and } |a| \leq N-1\} \sqcup \{\perp a\dagger : a \in A^* \text{ and } |a| < N-1\}.$$</td></tr>
<tr><td>2501.06662_EQ0006</td><td>8</td><td>0.996</td><td>\begin{aligned} & T(x)=\left\{y \in \operatorname{ob} \left(\mathrm{~L}_{x}\right): y \text { is an unfinished text of length } N\right. \text { or } \\ & \qquad y \text { is a finished text such that }|y| \leq N\} . \end{aligned}</td><td>$$\begin{aligned} & T(x)=\left\{y \in \operatorname{ob} \left(\mathrm{~L}_{x}\right): y \text { is an unfinished text of length } N\right. \text { or } \\ & \qquad y \text { is a finished text such that }|y| \leq N\} . \end{aligned}$$</td><td>$$\begin{aligned} T(x) = \{y \in \mathrm{ob}(\mathsf{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that } |y| \leq N\}. \end{aligned}$$</td></tr>
<tr><td>2501.06662_EQ0034</td><td>17</td><td>0.998</td><td>\operatorname{Mag}(t \mathcal{M}):=\sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) .</td><td>$$\operatorname{Mag}(t \mathcal{M}):=\sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{t}^{-1}(x, y) .$$</td><td>$$\operatorname{Mag}(t\mathcal{M}) := \sum_{x,y\in\mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,y).$$</td></tr>
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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>2501.06662_EQ0003</td><td>4</td><td>■0.998</td><td>\underbrace{\#(T(\perp))}_{\text {deterministic LM }} \leq \lim _{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text {maximally random LM }} .</td><td>$\underbrace{\#(T(\perp))}_{\text {deterministic LM }} \leq \lim _{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text {maximally random LM }} .$</td><td>$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t\to\infty} \operatorname{Mag}(t\mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}} .$</td></tr>
<tr><td>2501.06662_ EQ0054 (14)</td><td>22</td><td>■0.998</td><td>\operatorname{Mag}(t \mathcal{M})=\operatorname{rank} H_{0,0}(\mathcal{M})-\sum_{\ell \geq 0} q^{\ell} \operatorname{rank} H_{1, \ell}(\mathcal{M}) .</td><td>$\operatorname{Mag}(t \mathcal{M})=\operatorname{rank} H_{0,0}(\mathcal{M})-\sum_{\ell \geq 0} q^{\ell} \operatorname{rank} H_{1, \ell}(\mathcal{M}) .$</td><td>$\operatorname{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell\geq 0} q^{\ell} \operatorname{rank} H_{1,\ell}(\mathcal{M}).$</td></tr>
<tr><td>2501.06662_EQ0019</td><td>14</td><td>■0.999</td><td>\mu_{P}(s, u)= \begin{cases}1 & \text { if } s=u \\ -\sum_{s \leq t<u} \mu_{P}(s, t) & \text { if } s<u \\ 0 & \text { otherwise } .\end{cases}</td><td>$\mu_{P}(s, u)= \begin{cases}1 & \text { if } s=u \\ -\sum_{s \leq t<u} \mu_{P}(s, t) & \text { if } s<u \\ 0 & \text { otherwise } .\end{cases}$</td><td>$\mu_P(s,u) = \begin{cases} 1 & \text{if } s=u \\ -\sum\limits_{s\leq t<u} \mu_P(s,t) & \text{if } s<u \\ 0 & \text{otherwise.} \end{cases}$</td></tr>
<tr><td>2501.06662_EQ0042</td><td>19</td><td>■0.999</td><td>H\left(p_{x}\right)=-\left.\frac{\mathrm{d}}{\mathrm{~d} t} Z_{x}(t)\right|_{t=1},</td><td>$H\left(p_{x}\right)=-\left.\frac{\mathrm{d}}{\mathrm{~d} t} Z_{x}(t)\right|_{t=1},$</td><td>$H(p_x) = -\left.\frac{\mathrm{d}}{\mathrm{d}t} Z_x(t)\right|_{t=1},$</td></tr>
<tr><td>2501.06662_EQ0004</td><td>5</td><td>■1.000</td><td>\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} e^{-t \ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right),</td><td>$\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} e^{-t \ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right),$</td><td>$\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k\geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$</td></tr>
<tr><td>2501.06662_EQ0012</td><td>11</td><td>■1.000</td><td>\begin{aligned} & x=\perp a_{1} \cdots a_{t} \\ & y=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \\ & z=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \cdots a_{t+k+k^{\prime}} \end{aligned}</td><td>$\begin{aligned} & x=\perp a_{1} \cdots a_{t} \\ & y=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \\ & z=\perp a_{1} \cdots a_{t} \cdots a_{t+k} \cdots a_{t+k+k^{\prime}} \end{aligned}$</td><td>$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0031</td><td>17</td><td>■1.000</td><td>\begin{aligned} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} & =\pi\left(y_{1} \mid y_{0}\right)^{t} \pi\left(y_{2} \mid y_{1}\right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\pi\left(y_{2} \mid y_{0}\right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\quad \vdots \\ & =\pi\left(y_{k} \mid y_{0}\right)^{t} \end{aligned}</td><td>$\begin{aligned} \prod_{i=1}^{k} \pi\left(y_{i} \mid y_{i-1}\right)^{t} & =\pi\left(y_{1} \mid y_{0}\right)^{t} \pi\left(y_{2} \mid y_{1}\right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\pi\left(y_{2} \mid y_{0}\right)^{t} \pi\left(y_{3} \mid y_{2}\right)^{t} \cdots \pi\left(y_{k} \mid y_{k-1}\right)^{t} \\ & =\quad \vdots \\ & =\pi\left(y_{k} \mid y_{0}\right)^{t} \end{aligned}$</td><td>$\begin{aligned} \prod_{i=1}^{k} \pi(y_i \mid y_{i-1})^t &= \pi(y_1 \mid y_0)^t \pi(y_2 \mid y_1)^t \pi(y_3 \mid y_2)^t \cdots \pi(y_k \mid y_{k-1})^t \\ &= \pi(y_2 \mid y_0)^t \pi(y_3 \mid y_2)^t \cdots \pi(y_k \mid y_{k-1})^t \\ &= \quad \vdots \\ &= \pi(y_k \mid y_0)^t. \end{aligned}$</td></tr>
<tr><td>2501.06662_EQ0007</td><td>8</td><td>■1.000</td><td>p\left(a_{3} \mid \perp a\right) p\left(a_{4} \mid \perp a a_{3}\right) p\left(a_{5} \mid \perp a a_{3} a_{4}\right) p\left(\dagger \mid \perp a a_{3} a_{4} a_{5}\right) .</td><td>$p\left(a_{3} \mid \perp a\right) p\left(a_{4} \mid \perp a a_{3}\right) p\left(a_{5} \mid \perp a a_{3} a_{4}\right) p\left(\dagger \mid \perp a a_{3} a_{4} a_{5}\right) .$</td><td>$p(a_3 \mid \perp a) p(a_4 \mid \perp a a_3) p(a_5 \mid \perp a a_3 a_4) p(\dagger \mid \perp a a_3 a_4 a_5).$</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_ EQ0008 (1)	9	■1.000	`\pi(y \mid x):=\left\{\begin{array}{ll} 1 & \text { if } x=y \\ 0 & \text { if } x \nrightarrow y \\ \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) & \text { if } x \rightarrow y \end{array} .\right.`	$\pi(y \mid x):=\left\{\begin{array}{ll} 1 & \text { if } x=y \\ 0 & \text { if } x \nrightarrow y \\ \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) & \text { if } x \rightarrow y \end{array} .\right.$	$\pi(y\|x):=\begin{cases} 1 & \text{if } x=y \\ 0 & \text{if } x \nrightarrow y \\ \prod_{i=1}^{k} p(a_{t+i}\|y_{<t+i}) & \text{if } x \rightarrow y \end{cases}.$
2501.06662_ EQ0009 (2)	9	■1.000	`\sum_{y \in T(x)} \pi(y \mid x)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)=1,`	$\sum_{y \in T(x)} \pi(y \mid x)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)=1,$	$\sum_{y \in T(x)} \pi(y\|x)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)=1,$
2501.06662_ EQ0010 (4)	9	■1.000	`\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) & = \sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a \in A^{m}} \pi(x a \mid x) \\ & =\sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{\substack{a=a^{\prime} a^{\prime \prime} \\ a^{\prime} \in A^{m-1}, a^{\prime \prime} \in A}} p\left(a^{\prime \prime} \mid x a^{\prime}\right) \pi\left(x a^{\prime} \mid x\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \sum_{a^{\prime \prime} \in A \cup\{\dagger\}} p\left(a^{\prime \prime} \mid x a^{\prime}\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \end{aligned}`	$\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) & = \sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a \in A^{m}} \pi(x a \mid x) \\ & =\sum_{i=0}^{m-1} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{\substack{a=a^{\prime} a^{\prime \prime} \\ a^{\prime} \in A^{m-1}, a^{\prime \prime} \in A}} p\left(a^{\prime \prime} \mid x a^{\prime}\right) \pi\left(x a^{\prime} \mid x\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+\sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \sum_{a^{\prime \prime} \in A \cup\{\dagger\}} p\left(a^{\prime \prime} \mid x a^{\prime}\right) \\ & =\sum_{i=0}^{m-2} \sum_{a^{\prime} \in A^{i}} \pi\left(x a^{\prime} \dagger \mid x\right)+ \sum_{a^{\prime} \in A^{m-1}} \pi\left(x a^{\prime} \mid x\right) \end{aligned}$	$\begin{aligned} \sum_{y \in T(x)} \pi(y\|x) & = \sum_{i=0}^{m-1} \sum_{a' \in A^{i}} \pi(x a' \dagger \|x)+\sum_{a \in A^{m}} \pi(x a\|x) \\ & =\sum_{i=0}^{m-1} \sum_{a' \in A^{i}} \pi(x a' \dagger \|x)+ \sum_{\substack{a=a' a'' \\ a' \in A^{m-1}, a'' \in A}} p(a''\|x a') \pi(x a'\|x) \\ & =\sum_{i=0}^{m-2} \sum_{a' \in A^{i}} \pi(x a' \dagger \|x)+\sum_{a' \in A^{m-1}} \pi(x a'\|x) \sum_{a'' \in A \cup\{\dagger\}} p(a''\|x a') \\ & =\sum_{i=0}^{m-2} \sum_{a' \in A^{i}} \pi(x a' \dagger \|x)+ \sum_{a' \in A^{m-1}} \pi(x a'\|x) \end{aligned}$
2501.06662_EQ0011	10	■1.000	`\begin{aligned} 1 & \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) & \leq \mathcal{C}(x, z) \end{aligned}`	$\begin{aligned} 1 & \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) & \leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 & \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) & \leq \mathcal{C}(x, z) \end{aligned}$

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2501.06662_ EQ0013 (6)	11	■ 1.000	`\begin{aligned} \pi(y \mid x) \pi(z \mid y) & = \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) \prod_{j=1}^{k^{\prime}} p\left(a_{t+k+j} \mid z_{<t+k+j} \right) \\ & =\prod_{i=1}^{k+k^{\prime}} p\left(a_{t+i} \mid z_{<t+i}\right) \\ & =\pi(z \mid x) \end{aligned}`	$\begin{aligned} \pi(y \mid x) \pi(z \mid y) & = \prod_{i=1}^{k} p\left(a_{t+i} \mid y_{<t+i}\right) \prod_{j=1}^{k^{\prime}} p\left(a_{t+k+j} \mid z_{<t+k+j} \right) \\ & =\prod_{i=1}^{k+k^{\prime}} p\left(a_{t+i} \mid z_{<t+i}\right) \\ & =\pi(z \mid x) \end{aligned}$	$\begin{aligned} \pi(y\|x)\pi(z\|y) & = \prod_{i=1}^{k} p(a_{t+i}\|y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j}\|z_{<t+k+j}) \\ & = \prod_{i=1}^{k+k'} p(a_{t+i}\|z_{<t+i}) \\ & = \pi(z\|x). \end{aligned}$
2501.06662_EQ0014	12	■ 1.000	`d(x, y):=-\ln \pi(y \mid x) .`	$d(x, y):=-\ln \pi(y \mid x) .$	$d(x,y) := -\ln \pi(y\|x).$
2501.06662_EQ0016	13	■ 1.000	`(f * g)(s, u):=\sum_{s \leq t \leq u} f(s, t) g(t, u),`	$(f * g)(s, u):=\sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s,u) := \sum_{s \leq t \leq u} f(s,t)g(t,u),$
2501.06662_EQ0017	13	■ 1.000	`\delta(s, u)= \begin{cases}1 & \text { if } s=u \\ 0 & \text { if } s \neq u\end{cases}`	$\delta(s, u)= \begin{cases}1 & \text { if } s=u \\ 0 & \text { if } s \neq u\end{cases}$	$\delta(s,u) = \begin{cases}1 & \text{if } s=u \\ 0 & \text{if } s \neq u\end{cases}$
2501.06662_EQ0018	13	■ 1.000	`\zeta_{P}(s, u)=1, \quad \text { for all } s \leq u \text { in } P .`	$\zeta_{P}(s, u)=1, \quad \text { for all } s \leq u \text { in } P .$	$\zeta_P(s,u) = 1, \quad \text{for all } s \leq u \text{ in } P.$
2501.06662_EQ0020	14	■ 1.000	`\begin{aligned} \mu_{\mathrm{L}}(x, x) & =1 \\ \mu_{\mathrm{L}}\left(x, x a_{1}\right) & =-\mu_{ \mathrm{L}}(x, x)=-1 \\ \mu_{\mathrm{L}}\left(x, x a_{1} a_{2}\right) & =-\mu_{\mathrm{L}}(x, x)-\mu_{ \mathrm{L}}\left(x, x a_{1}\right)=-1+1=0 \\ \mu_{ \mathrm{L}}\left(x, x a_{1} a_{2} a_{3}\right) & =- \mu_{\mathrm{L}}(x, x)-\mu_{\mathrm{L}}\left(x, x a_{1}\right)-\mu_{\mathrm{L}}\left(x, x a_{1} a_{2} \right)=-1+1+0=0 . \end{aligned}`	$\begin{aligned} \mu_{\mathrm{L}}(x, x) & =1 \\ \mu_{\mathrm{L}}\left(x, x a_{1}\right) & =-\mu_{\mathrm{L}}(x, x)=-1 \\ \mu_{\mathrm{L}}\left(x, x a_{1} a_{2}\right) & =-\mu_{\mathrm{L}}(x, x)-\mu_{\mathrm{L}}\left(x, x a_{1}\right)=-1+1=0 \\ \mu_{\mathrm{L}}\left(x, x a_{1} a_{2} a_{3}\right) & =-\mu_{\mathrm{L}}(x, x)-\mu_{\mathrm{L}}\left(x, x a_{1}\right)-\mu_{\mathrm{L}}\left(x, x a_{1} a_{2}\right)=-1+1+0=0 . \end{aligned}$	$\begin{aligned} \mu_{\mathsf{L}}(x,x) & = 1 \\ \mu_{\mathsf{L}}(x,xa_1) & = -\mu_{\mathsf{L}}(x,x) = -1 \\ \mu_{\mathsf{L}}(x,xa_1a_2) & = -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) = -1+1 = 0 \\ \mu_{\mathsf{L}}(x,xa_1a_2a_3) & = -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) - \mu_{\mathsf{L}}(x,xa_1a_2) = -1+1+0 = 0. \end{aligned}$
2501.06662_EQ0024	15	■ 1.000	`\zeta_{\mathcal{M}}(x, y):=e^{-d(x, y)}=\pi(y \mid x),`	$\zeta_{\mathcal{M}}(x, y):=e^{-d(x, y)}=\pi(y \mid x),$	$\zeta_{\mathcal{M}}(x,y) := e^{-d(x,y)} = \pi(y\|x),$
2501.06662_EQ0025	15	■ 1.000	`\left(\zeta_{\mathcal{M}}\right)_{t}(x, y):=e^{-t d(x, y)}=\pi(y \mid x)^{t} .`	$\left(\zeta_{\mathcal{M}}\right)_{t}(x, y):=e^{-t d(x, y)}=\pi(y \mid x)^{t} .$	$(\zeta_{\mathcal{M}})_t(x,y) := e^{-td(x,y)} = \pi(y\|x)^t.$
2501.06662_ EQ0027 (9)	16	■ 1.000	`\zeta_{t}^{-1}=\sum_{k \geq 0}(-1)^{k}\left(\zeta_{t}-\delta\right)^{k}`	$\zeta_{t}^{-1}=\sum_{k \geq 0}(-1)^{k}\left(\zeta_{t}-\delta\right)^{k}$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k,$
2501.06662_EQ0028	16	■ 1.000	`\left(\zeta_{t}-\delta\right)(x, y)=\pi(x, y)^{t} \mathbf{1}_{(x, y) \in E},`	$\left(\zeta_{t}-\delta\right)(x, y)=\pi(x, y)^{t} \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x,y) = \pi(x,y)^t \mathbf{1}_{(x,y) \in E},$
2501.06662_EQ0030	16	■ 1.000	`\zeta_{t}^{-1}(x, y)=\pi(y \mid x)^{t} \zeta_{ \mathrm{L}}^{-1}(x, y) .`	$\zeta_{t}^{-1}(x, y)=\pi(y \mid x)^{t} \zeta_{\mathrm{L}}^{-1}(x, y) .$	$\zeta_t^{-1}(x,y) = \pi(y\|x)^t\, \zeta_{\mathsf{L}}^{-1}(x,y).$
2501.06662_EQ0035	17	■ 1.000	`H_{t}(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^{n} p_{i}^{t} \right) .`	$H_{t}(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^{n} p_{i}^{t}\right) .$	$H_t(p) = \frac{1}{t-1}\left(1 - \sum_{i=1}^{n} p_i^t\right).$

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2501.06662_EQ0038	18	■1.000	`\lim _{t \rightarrow 1} H_{t}(p)=\lim _{t \rightarrow 1} \frac{1}{t-1}\left(1-\sum_{s \in S} p(s)^{t}\right)=-\sum_{s \in S} p(s) \ln p(s)=: H(p) .`	$\lim _{t \rightarrow 1} H_{t}(p)=\lim _{t \rightarrow 1} \frac{1}{t-1}\left(1-\sum_{s \in S} p(s)^{t}\right)=-\sum_{s \in S} p(s) \ln p(s)=: H(p) .$	$\lim_{t\to 1} H_t(p)=\lim_{t\to 1}\frac{1}{t-1}\left(1-\sum_{s\in S}p(s)^t\right)=-\sum_{s\in S}p(s)\ln p(s)=:H(p).$
2501.06662_EQ0041	19	■1.000	`Z_{x}(t)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}=\sum_{a \in A \cup\{\dagger\}} e^{-t\left(-\ln p_{x}(a)\right)}`	$Z_{x}(t)=\sum_{a \in A \cup\{\dagger\}} p_{x}(a)^{t}=\sum_{a \in A \cup\{\dagger\}} e^{-t\left(-\ln p_{x}(a)\right)}$	$Z_x(t)=\sum_{a\in A\cup\{\dagger\}} p_x(a)^t=\sum_{a\in A\cup\{\dagger\}} e^{-t(-\ln p_x(a))}$
2501.06662_EQ0043	19	■1.000	`\mathbb{E}_{\rho}(E)=\sum_{s \in S} E(s) \frac{e^{-t E(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{~d} t} \ln \tilde{Z}(t)=\frac{f^{\prime}(t)}{\tilde{Z}(t)},`	$\mathbb{E}_{\rho}(E)=\sum_{s \in S} E(s) \frac{e^{-t E(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{~d} t} \ln \tilde{Z}(t)=\frac{f^{\prime}(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E)=\sum_{s\in S}E(s)\frac{e^{-tE(s)}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{d}t}\ln\tilde{Z}(t)=\frac{f'(t)}{\tilde{Z}(t)},$
2501.06662_EQ0047	20	■1.000	`G_{k, \ell}=\left\{\left(y_{0}, \ldots, y_{k}\right) \in M^{k+1} \mid \sum_{i=0}^{k-1} d\left(y_{i}, y_{i+1}\right)=\ell \text { and for all } i, y_{i} \neq y_{i+1}\right\} .`	$G_{k, \ell}=\left\{\left(y_{0}, \ldots, y_{k}\right) \in M^{k+1} \mid \sum_{i=0}^{k-1} d\left(y_{i}, y_{i+1}\right)=\ell \text { and for all } i, y_{i} \neq y_{i+1}\right\} .$	$G_{k,\ell}=\left\{(y_0,\ldots,y_k)\in M^{k+1} \mid \sum_{i=0}^{k-1}d(y_i,y_{i+1})=\ell \text{ and for all } i,\ y_i\neq y_{i+1}\right\}.$
2501.06662_EQ0048	20	■1.000	`\partial_{k}=\sum_{i=0}^{k}(-1)^{i} \partial_{k}^{i}`	$\partial_{k}=\sum_{i=0}^{k}(-1)^{i} \partial_{k}^{i}$	$\partial_k=\sum_{i=0}^{k}(-1)^i\partial_k^i$
2501.06662_EQ0050	21	■1.000	`\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} q^{\ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right)`	$\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} q^{\ell} \sum_{k \geq 0}(-1)^{k} \operatorname{rank}\left(H_{k, \ell}(\mathcal{M})\right)$	$\operatorname{Mag}(t\mathcal{M})=\sum_{\ell}q^\ell\sum_{k\geq 0}(-1)^k\operatorname{rank}(H_{k,\ell}(\mathcal{M}))$
2501.06662_EQ0053	21	■1.000	`\sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}=(-1)^{k} q^{\ell} \# G_{k, \ell}=(-1)^{k} q^{\ell} \operatorname{rank}\left(M C_{k, \ell}(\mathcal{M})\right) .`	$\sum_{c \in G_{k, \ell}}(-1)^{k} e^{-t \ell}=(-1)^{k} q^{\ell} \# G_{k, \ell}=(-1)^{k} q^{\ell} \operatorname{rank}\left(M C_{k, \ell}(\mathcal{M})\right) .$	$\sum_{c\in G_{k,\ell}}(-1)^k e^{-t\ell}=(-1)^k q^\ell \# G_{k,\ell}=(-1)^k q^\ell\operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$
2501.06662_EQ0055	22	■1.000	`P P L(x)=\exp \left\{-\frac{1}{n} \sum_{i=1}^{n} \ln p \left(a_{i} \mid y_{<i}\right)\right\}`	$P P L(x)=\exp \left\{-\frac{1}{n} \sum_{i=1}^{n} \ln p \left(a_{i} \mid y_{<i}\right)\right\}$	$PPL(x)=\exp\left\{-\frac{1}{n}\sum_{i=1}^{n}\ln p(a_i \vert y_{<i})\right\}$
2501.06662_EQ0057	23	■1.000	`\theta(a, b)=0 \quad \text { if } \quad \mathrm{A}(a, b)=\varnothing .`	$\theta(a, b)=0 \quad \text { if } \quad \mathrm{A}(a, b)=\varnothing .$	$\theta(a,b)=0 \quad \text{if} \quad \mathsf{A}(a,b)=\varnothing.$
2501.06662_EQ0060	23	■1.000	`-\sum_{a \in A} \sum_{a^{\prime} \in A} p_{\perp}(a) P \left(a, a^{\prime}\right) \log P\left(a, a^{\prime}\right),`	$-\sum_{a \in A} \sum_{a^{\prime} \in A} p_{\perp}(a) P \left(a, a^{\prime}\right) \log P\left(a, a^{\prime}\right),$	$-\sum_{a\in A}\sum_{a'\in A}p_\perp(a)P(a,a')\log P(a,a'),$